

# Revision

Bootstrapping - used when less data; mimics resampling  
drawing repeated sample with replacement

Point Estimate -

- a single number; It mostly tell sample mean
- BEST GUESS for POP. mean
- Need something to estimate accuracy of guess

Confidence level - How sure we are that our <sup>Point Estimate</sup> guess is correct

Significance level - The risk we are willing to take that it is wrong

1 - confidence level

Confidence Interval :- Range around point estimate (guess) where true value lie

$$\text{Confidence Interval (CI)} = \text{Point Estimate} \pm \text{Margin of error (MOE)}$$
$$\bar{x} \pm z \cdot \frac{s}{\sqrt{n}} \quad \text{Confidence Interval}$$

z value

2. MOE = CI

Hypothesis — A statement we want to test  
→

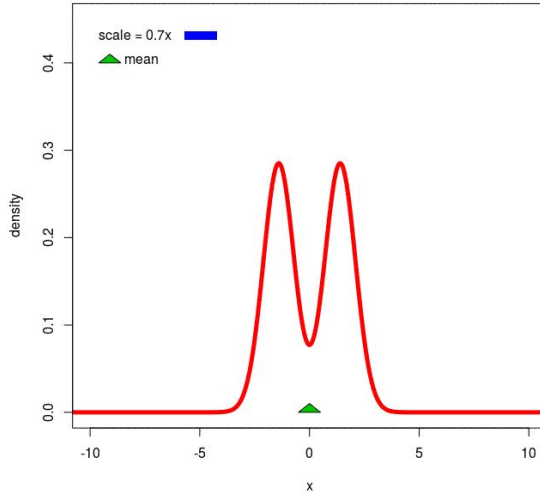
eg:- Samparan Sir R

1 glass - 150/- juice sell  
Sat Sun ]  
100 120

- Sat Sun Mon Tues  
100 120 140 160 120  
Tra R R R R

Assumptions

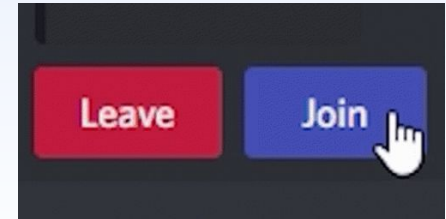
- Red cup will increase my sale ] — ALTERNATE HYPOTHESIS
- No effect on sale by Red cup ] — NULL HYPOTHESIS  
used



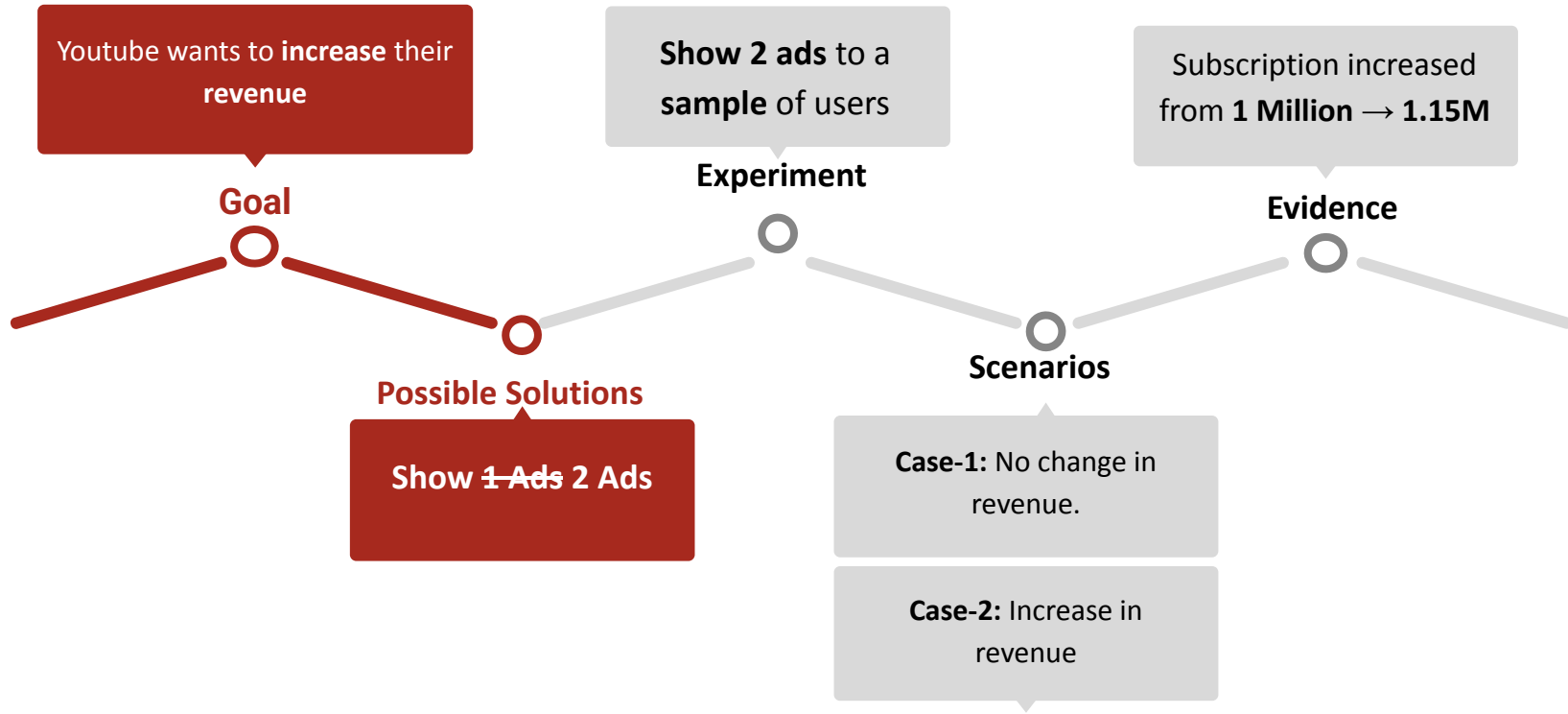
## Module 4 – Inferential Statistics

# Intro to Hypothesis Testing

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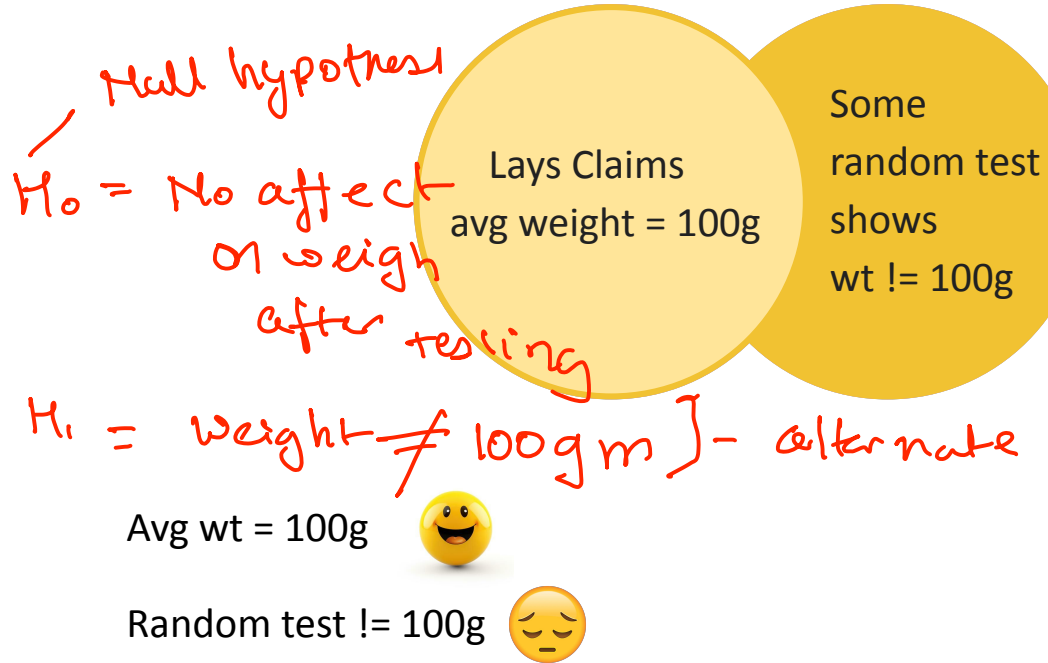


# Hypothesis Testing – Youtube Example



Can we **confidently** say that **showing ads leads to increase in revenue?**

# Hypothesis Testing – Lays Example



How many samples of Lays packets will you test to be confident?

# Hypothesis Testing – Courtroom Example

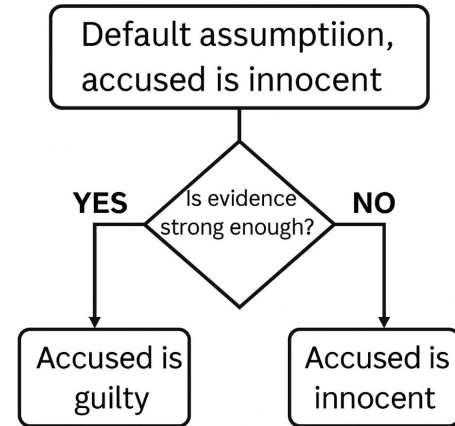


Person is brought to court as a suspect for their hearing.

- Is the accused guilty?
- **Default assumption** → Innocent until proven guilty.
- **Need evidence** to make decision

*Null hypothesis — He is innocent after seeing evidence*

*Alternate hypothesis — Guilty*



How many witnesses you will require to prove the accused guilty?

# Understanding Hypotheses

## Hypothesis:

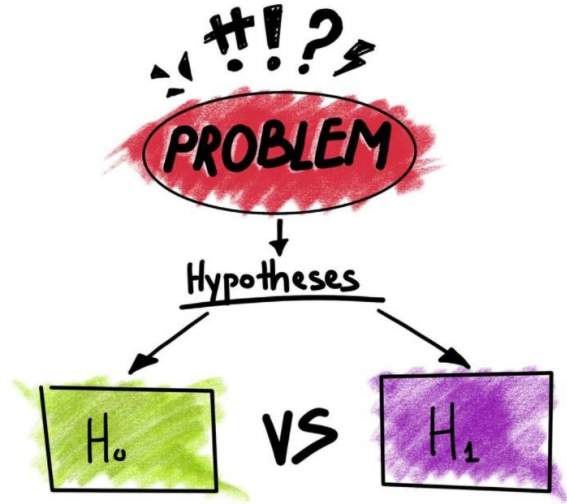
A hypothesis is a **statement or claim** about a population parameter that we want to **test using sample data**.

## Null Hypothesis ( $H_0$ ):

It is the **default assumption** that there is **no effect** or no difference.

## Alternative Hypothesis ( $H_1$ ):

It is the **statement we want to prove** — that there is an effect or a change has occurred.



# Null & Alternate Hypotheses



## Case-1

Youtube Subscriptions

$H_0$ : No of subscriptions = 1M

$H_1$ : No of subscriptions > 1M

## Case-2

Average Lays Weight

$H_0$ : Average weight = 100g

$H_1$ : Average weight  $\neq$  100g

## Case-3

Courtroom Case

$H_0$ : Accused is innocent

$H_1$ : Accused is guilty

$H_0$  - Null hypothesis ( $H_0$ )  
 $H_1$  - Alternate

# Hypothesis Testing – An end-to-end case study

# Productivity After Training

A **car company** wants to evaluate if a new **training program** has improved the **productivity** of its employees.

## Before Training:

- Average productivity = **50 cars/day**
- Population standard deviation  $\sigma=5$

## After Training:

- A sample of **30 employees** was *well selected to represent the population*
- Sample average productivity = **53 cars/day**



Null hypot  
Alter hypotn

$H_0$ : No change in productivity  $\rightarrow \mu = 50$   
 $H_1$ : Increase in productivity  $\rightarrow \mu > 50$

sample  
data

after training

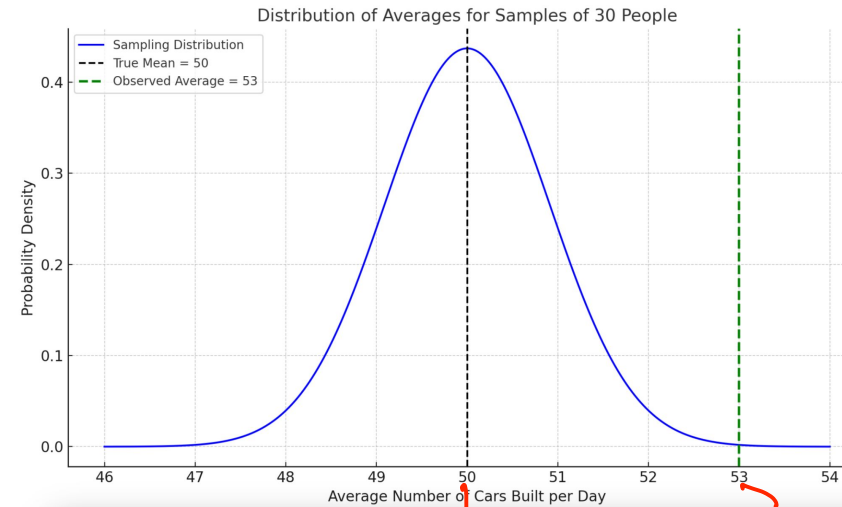
# Distribution of Sample Means

We have taken a **single sample** and we claim that the productivity has **increased from 50 to 53**.

What Happens When We Take **Many Random Samples**?

**It Follows a Normal (Bell-Shaped) Distribution!**

- When we collect averages from many random groups of 30 employees, those averages tend to **follow a bell curve** (called a **normal distribution**).



before training  
50 cars per day

after training

# Understanding Variability in Averages

**Most group averages will be close to 50**

- If we take many random groups of 30 employees, most of their **average production** will be somewhere around **50**.
- This is what the **Central Limit Theorem** says.

**But sometimes, just by chance, we might get 51 or 52**

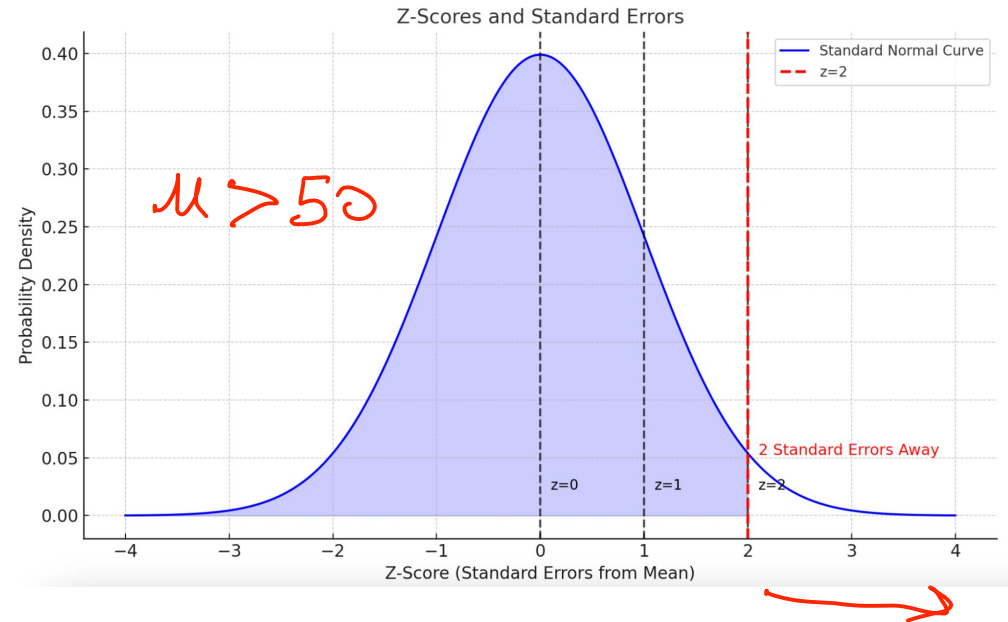
- That's **still okay**.
- Small bumps can happen **naturally**, even when nothing has changed.
- But **how far is too far to still call it “just by chance”?**



# Converting to Z-Score

First we convert our sample mean into a **z-score**, which says “How many standard errors away from 50?”

A **z-score** measures the distance between your observed mean and the hypothesized mean, in units of “standard error.”



# Setting the Cut-Off

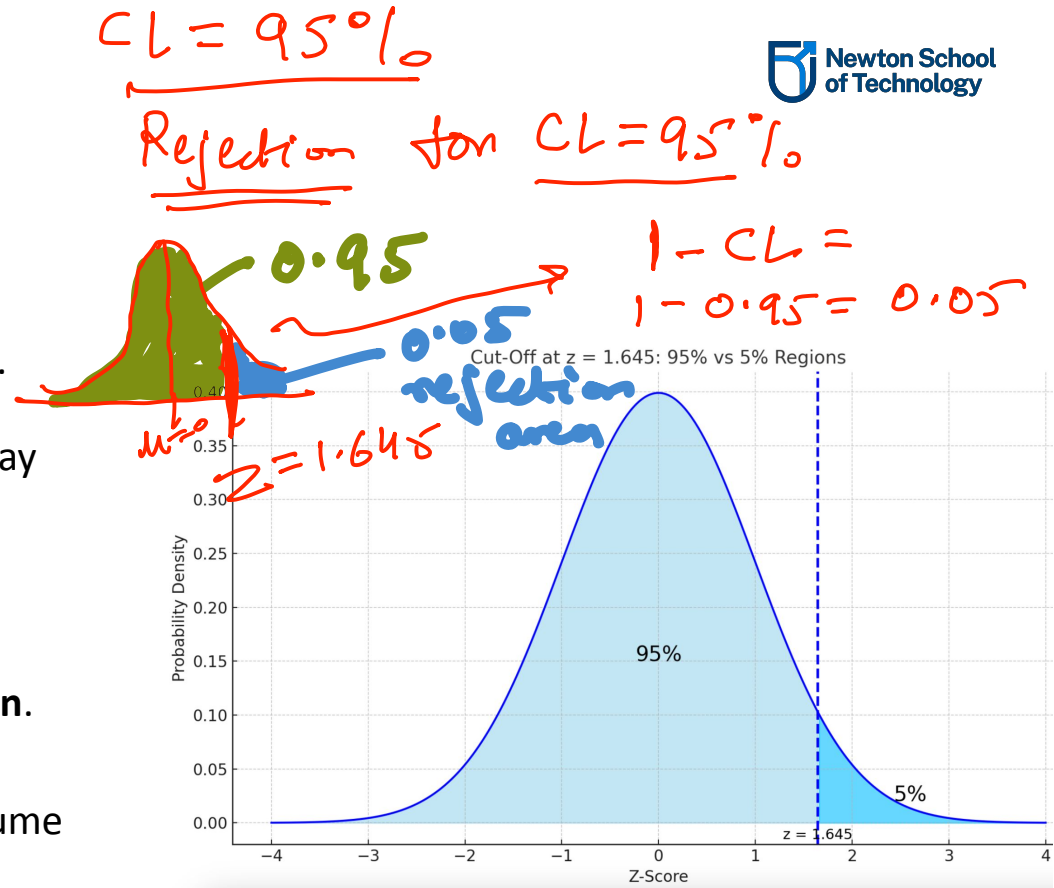
We now set a **cut-off value** to decide if the result is too far to be random.

We **compare the z-score to a cut-off point**.

If the **z-score exceeds** a certain cutoff we say it's "**too far**" to be random.

**Who decides this cut-off?**

- It's usually **pre-decided by convention**.
- In most real-world problems, we assume it as **5%**.



# Identifying the Rejection Threshold

We are told that the cut-off is **5%**, but now we need to find out:

**"What is the z-score at which 5% of values lie to the right?"**

We want to mark the point on the curve where only **5% of results fall above it.**

This means we are looking for the z-score such that :  **$P(Z > z) = 0.05$**

We look in the z-table to find the value of z where :  **$P(Z < z) = 0.95$**

This is because the left area must be 95%, so the right area (tail) is the remaining 5%

**$z = 1.645$  -> This is the cut-off point**

# Rejection Region

The **red dashed line at  $z = 1.645$**  is our **cutoff point**.

The area to the right of  $z = 1.645$  is called the **rejection region**.

If our  $z$ -value lies in that region, we **reject the null hypothesis**.

If we land there, we **no longer believe the old average of 50**.



$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \rightarrow \frac{(\bar{x}) - \mu - (50)}{\frac{\sigma}{\sqrt{n}}} \quad \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = z$$

mean of

# Interpreting the Z-Score

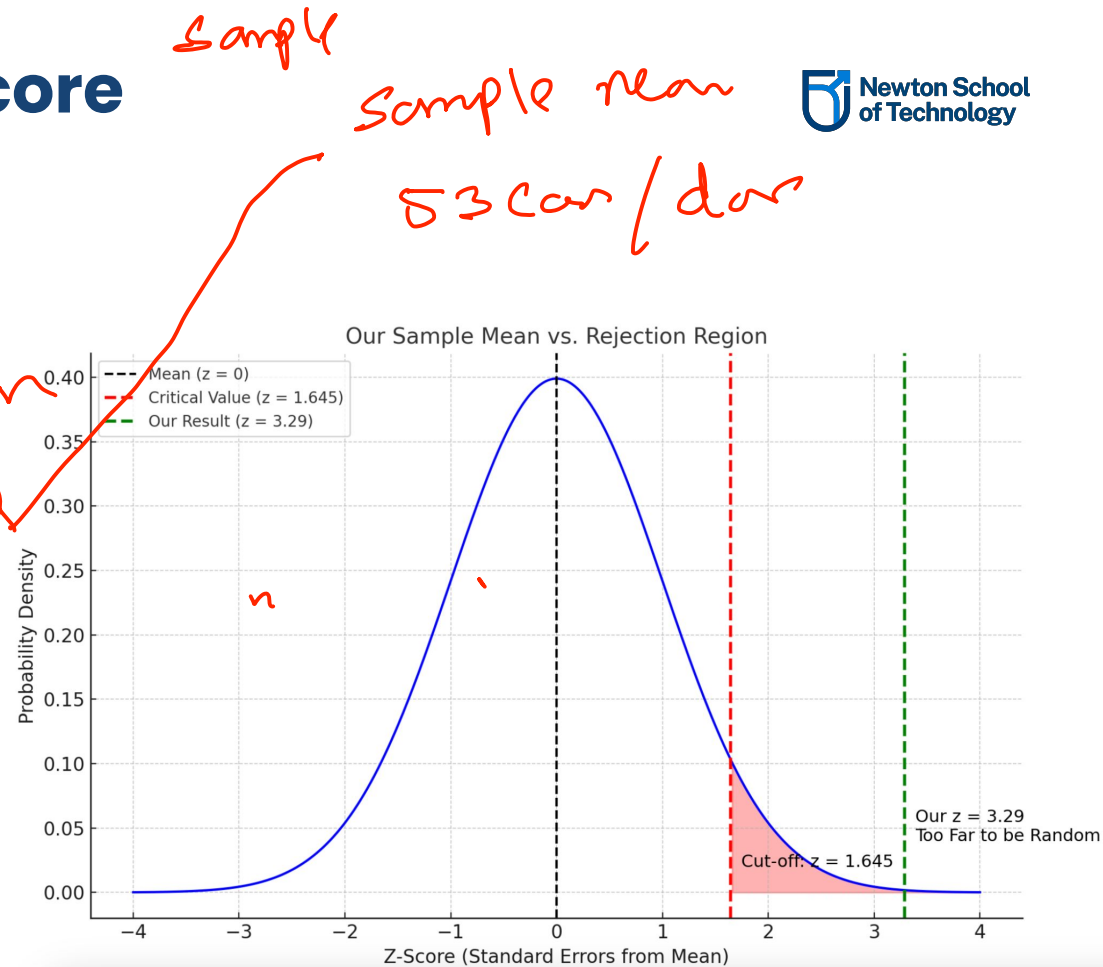
To check how far 53 is from 50 (in standard error units), we use:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$z = \frac{53 - 50}{5 / \sqrt{30}} = \frac{3}{0.913} \approx 3.29$$

Our result is 3.29 standard units above 50.  
That's way beyond the limit of 1.645.

Hence we fail to accept Null Hypothesis.



$H_0 - \mu = 50$  still

50 Car is still being man. after  
train

$H_1 = \mu > 50$

53 Car/day after training

**Let's try solving another question**

# Testing the Validity of Lays Packet Weight

## Assume Company Claim:

The average weight of a Lays packet is **50g**.

## Your Test:

- You randomly select a **sample of 40 packets**
- The **sample average weight** is **49g** — 
- The **population standard deviation** is known: **4g**

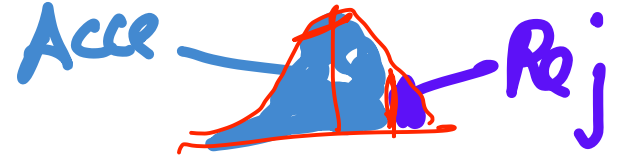


## What Are We Trying to Find?

We want to check if the **actual average weight differs** from the claimed 50g.

# Testing the Validity of Lays Packet Weight

- **Null Hypothesis ( $H_0$ ):**  
The average weight is 50g  $\rightarrow \mu = 50$
- **Alternate Hypothesis ( $H_1$ ):**  
The average weight is **different from 50g**  $\rightarrow \mu \neq 50$



$$\underline{\mu > 50}$$

We assume our cut-off value to be 5%

Sample data later

Will my cut-off point be 1.67?

# Testing the Validity of Lays Packet Weight

We are testing whether the **average weight** is **different** from 50g, not whether it is **more** or **less** than 50g specifically.

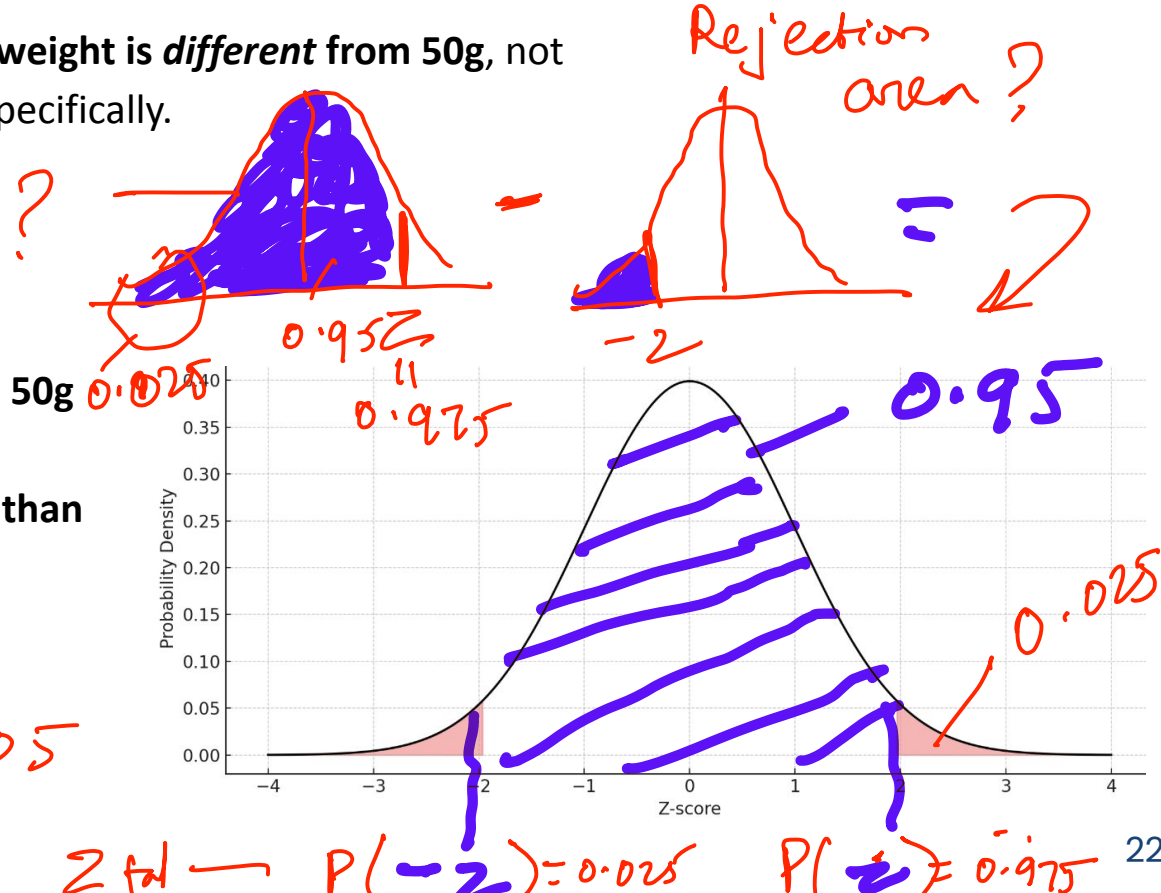
That means we are open to **both possibilities**:

- The average might be **less than 50g**
- OR the average might be **more than 50g**

$$CL = 95\%$$

$$\text{Sig lev} = 1 - 0.95 = 0.05$$

Rejection



# Testing the Validity of Lays Packet Weight

So we **care about both sides** of the normal curve — the **left tail** (too small) and the **right tail** (too large).

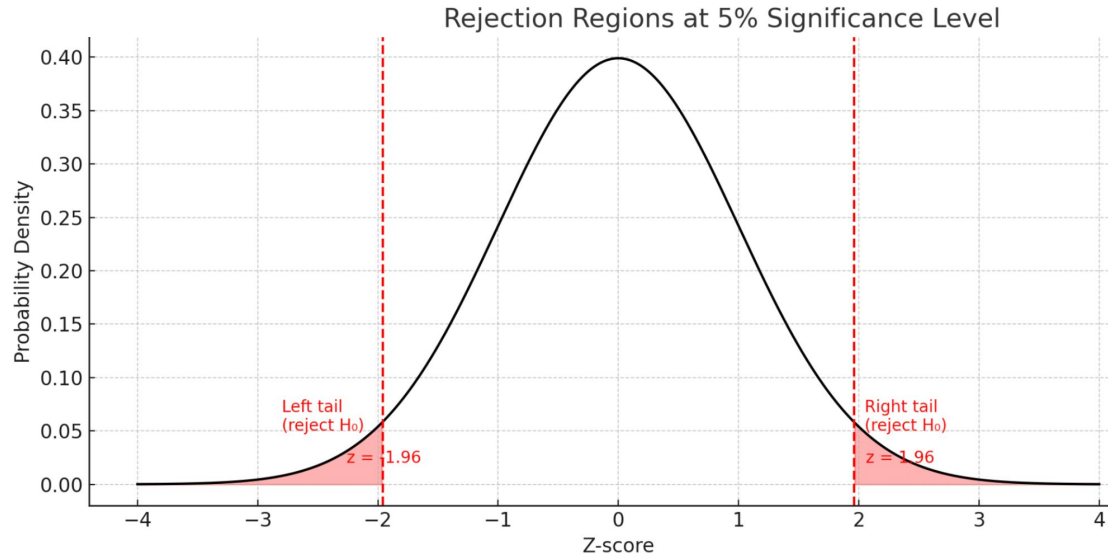
We still want to allow only a **5% error**, but now we **split it** into two equal parts:

- 2.5% on the **left**
- 2.5% on the **right**

**So the cut-off points become:**

**$z = -1.96$  on the left**

**$z = +1.96$  on the right**



# Testing the Validity of Lays Packet Weight

If the z-score is:

- Less than -1.96 OR
- More than +1.96

$$\bar{x} = 49 \text{ gm}$$
$$\mu = 50 \text{ gm}$$

→ We say: "This is too far from 50g to be random."

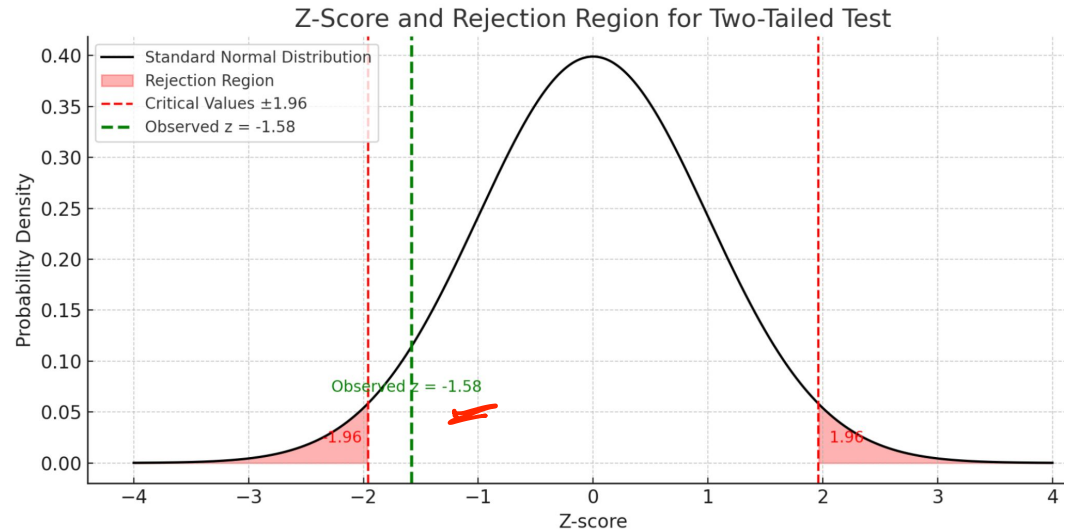
The **z-score of** sample mean (49g) is

$$z = \frac{49 - 50}{4/\sqrt{40}} = \frac{-1}{4/6.32} = \frac{-1}{0.632} \approx -1.58$$

# Testing the Validity of Lays Packet Weight

Since  $-1.96 < -1.58 < +1.96$ ,  
the z-score lies **within the acceptance region**,  
so we **do not reject the null hypothesis**.

**Conclusion:** There is **not enough evidence** to say the average weight is different from 50g.

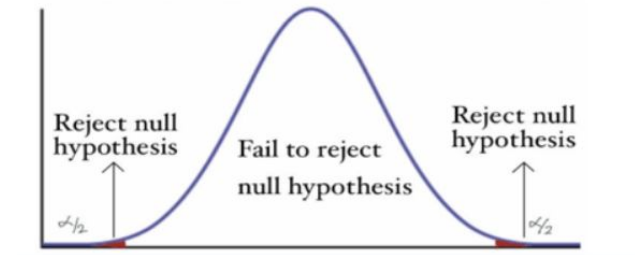
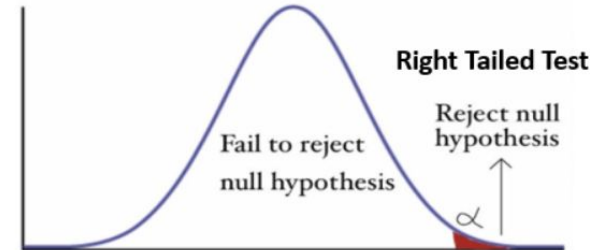
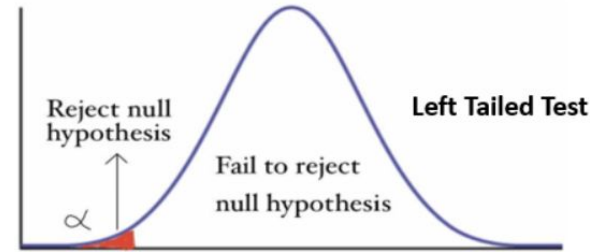


# Tailed Test

Depending on how we ask our question — whether we're only checking if the value is **more**, **less**, or **just different in any way** — we choose a different type of test.

A **one-tailed test** (right-tailed or left-tailed) is used if the alternative hypothesis specifically claims the parameter is **greater(right)** or **less(left)** than the hypothesized value.

A **two-tailed test** is used if the alternative hypothesis states that the parameter is **different** (could be higher or lower) than the hypothesized value.



# Burger Weight Claim

A burger place claims that each burger weighs **200g**. A customer who still feels hungry suspects the burgers are **lighter** than 200g.

What will be  $H_0$  and  $H_1$ ?

**Null Hypothesis ( $H_0$ ):** Average burger weight is 200g ( $\mu = 200$ ).

**Alternative Hypothesis ( $H_a$ ):** Average burger weight is less than 200g ( $\mu < 200$ ).

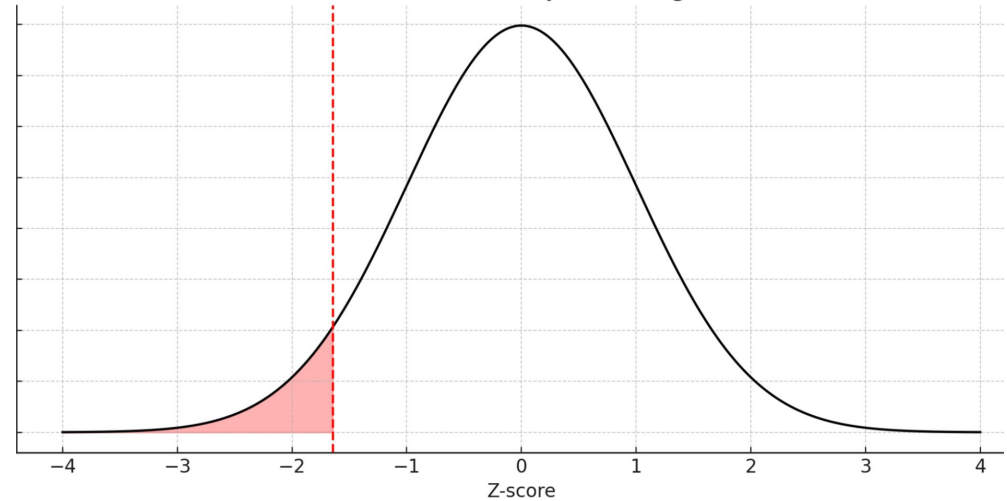


# Burger Weight Claim

Which tail will the test be for burger claim?

Since  $H_1$  states the mean is less than 200g, the evidence to reject  $H_0$  would appear in the **left tail** of the distribution.

Thus, this is a **left-tailed test**.



# Metal Rods Claim

A factory claims that the average strength of its metal rods is **at most 50 kg**. An engineer suspects the rods are actually **stronger** on average.

What will be  $H_0$  and  $H_1$ ?

**Null Hypothesis ( $H_0$ ):** The average strength is at most 50 kg ( $\mu \leq 50$ )

**Alternative Hypothesis ( $H_1$ ):** The average strength is more than 50 kg ( $\mu > 50$ )

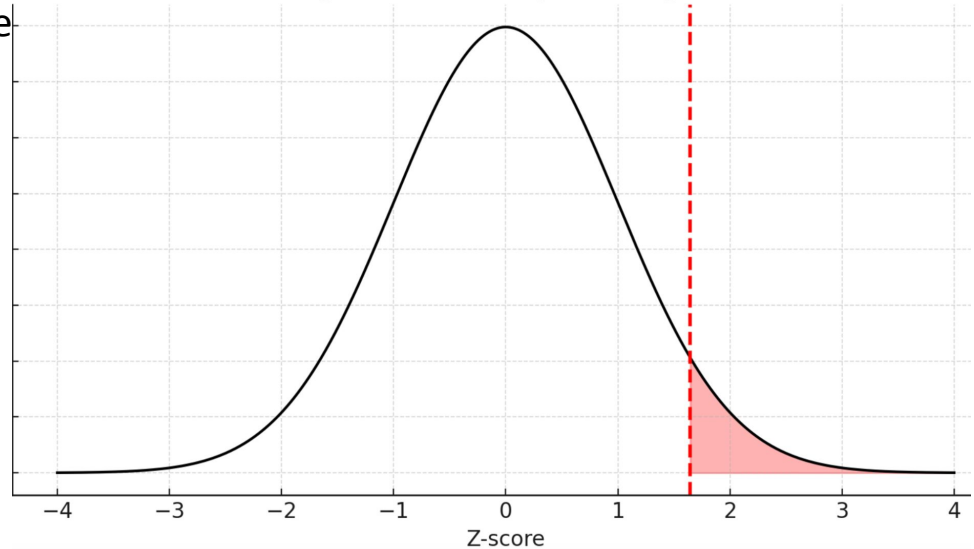


# Metal Rods Claim

Which tail will the test be?

Since  $H_1$  states the mean is greater than 50 kg,  
the evidence to reject  $H_0$  would appear in the **right tail** of the distribution.

Thus, this is a **right-tailed test**.



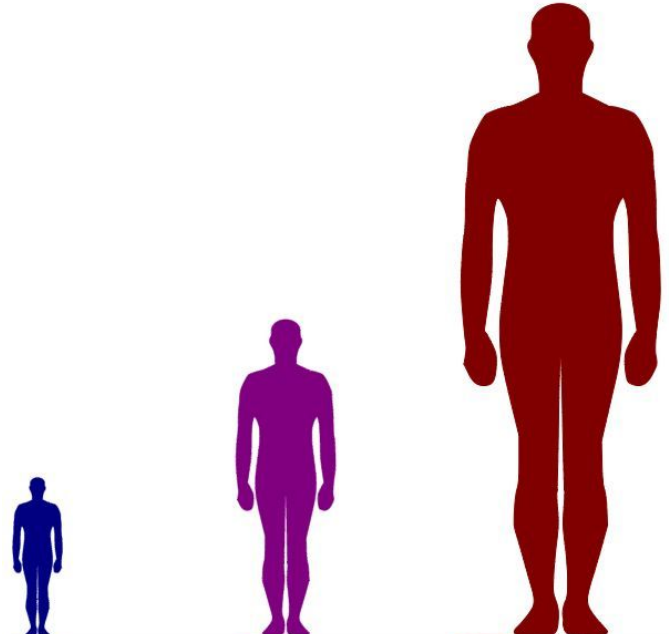
# Average Height Claim

Suppose you are looking at the height of people in India. It is believed that the average height of Indians is 65 inches. You want to find out if that holds true for the people of your state. Are they taller or shorter?

What will be  $H_0$  and  $H_1$ ?

**Null Hypothesis ( $H_0$ ):** The average height is 65 inches  
( $\mu = 65$ )

**Alternative Hypothesis ( $H_1$ ):** The average height is  
different from 65 inches ( $\mu \neq 65$ )

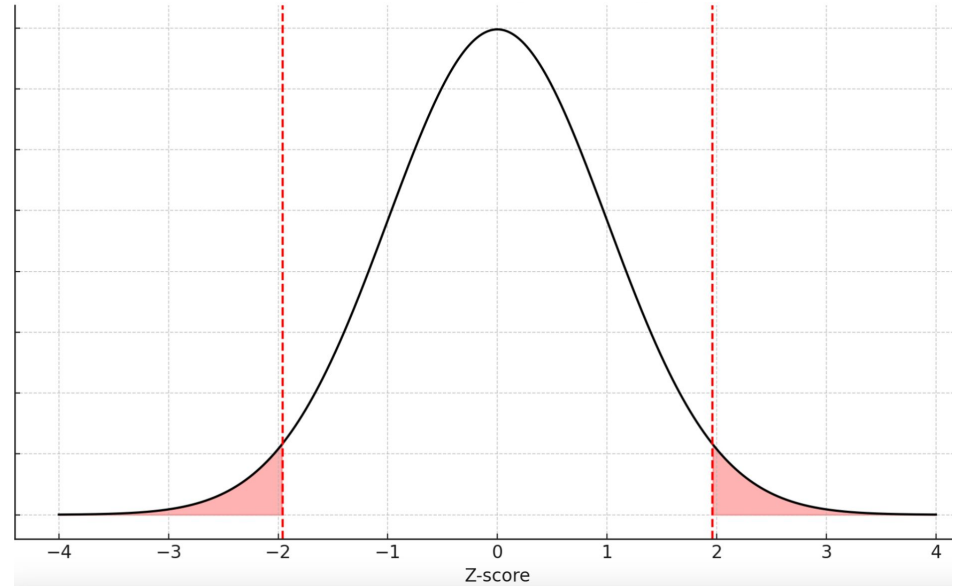


# Average Height Claim

Which tail will the test be?

Since  $H_1$  states the mean is **not equal to 65**, the evidence to reject  $H_0$  could appear in **both tails** of the distribution.

Thus, this is a **two-tailed test**.



# Hypothesis Testing: Rejection Region Method

## Rejection Region Approach – Step-by-Step Process

1. **Formulate the Null Hypothesis ( $H_0$ )**  
Define the claim to be tested (usually the "no effect" or "status quo").
2. **Select the Cut-off value = Significance Level ( $\alpha$ )**  
Common choices: 0.05, 0.01, 0.10
3. **Check Assumptions**  
Verify conditions for the chosen test (e.g., normality, independence, sample size).
4. **Decide Which Test to Perform**  
Choose the appropriate statistical test based on the scenario and data type.
5. **State the Test Statistic**  
Write the formula and logic for calculating the test statistic.
6. **Conduct the Test**  
Compute the value of the test statistic using the sample data.
7. **Make a Decision**

# Hypothesis Testing: Rejection Region Method

## 1. Set Hypotheses

- $H_0$ : No change/effect
- $H_1$ : There *is* a change/effect

## 2. Choose Significance Level ( $\alpha$ )

- Commonly **0.05 (5%)**
- Risk you're willing to take in rejecting  $H_0$

## 3. Compute Test Statistic

- Example: Z-score

## 4. Find Critical Value

- From **Z-table** based on  $\alpha$  i.e.  
(Significance level/ Cut-off value)

## 5. Define Rejection Region

- Area **beyond critical value**
- If test statistic lies here → **Reject  $H_0$**
- Else → **Fail to Reject  $H_0$**

# Key Takeaways

## 1. What is Hypothesis Testing?

A structured method to test claims about a population parameter using sample data.

- Null Hypothesis ( $H_0$ ): No change/effect
- Alternative Hypothesis ( $H_1$ ): There is a change/effect

## 2. Z-score

Z-score measures **how far** the sample mean is from the population mean in terms of **standard errors**. Helps quantify: *“Is this change real or just due to chance?”*

## 3. Rejection Region

- Area **beyond critical value** ( $z = \pm 1.96$  or  $z = 1.645$ ) is the rejection region.
- If test statistic falls in this area → **Reject  $H_0$**
- Otherwise → **Fail to Reject  $H_0$**

## 4. Rejection Region Method – Step-by-Step

1. **Set Hypotheses** –  $H_0$  and  $H_1$ .
2. **Choose Significance Level ( $\alpha$ )** – Commonly 0.05.
3. **Check Assumptions** – Normal distribution, known  $\sigma$ , independent sample.
4. **Compute Test Statistic (Z)** – Using sample data.
5. **Find Critical Z-value** – From Z-table using  $\alpha$ .
6. **Compare** – Does the Z fall in the rejection region?
7. **Make Decision** – Reject or Fail to Reject  $H_0$ .

## 5. Types of Tests: One-Tailed vs Two-Tailed

- **Right-Tailed Test** → Claim is “greater than”
- **Left-Tailed Test** → Claim is “less than”
- **Two-Tailed Test** → Claim is “different ( $\neq$ )”  
Use depends on **how the alternative hypothesis is framed**.

Practice questions- [Here](#), Q1 to Q13

**See You Guys  
in Next  
Session :)**